COMP1015: Programming Fundamentals and Algorithmic Design

ASSESSMENT TASK 2 — CASE STUDY/ANALYSIS TEMPLATE

Complete your personal details in the table below, then provide your submission below for all three sections. You must complete all details.

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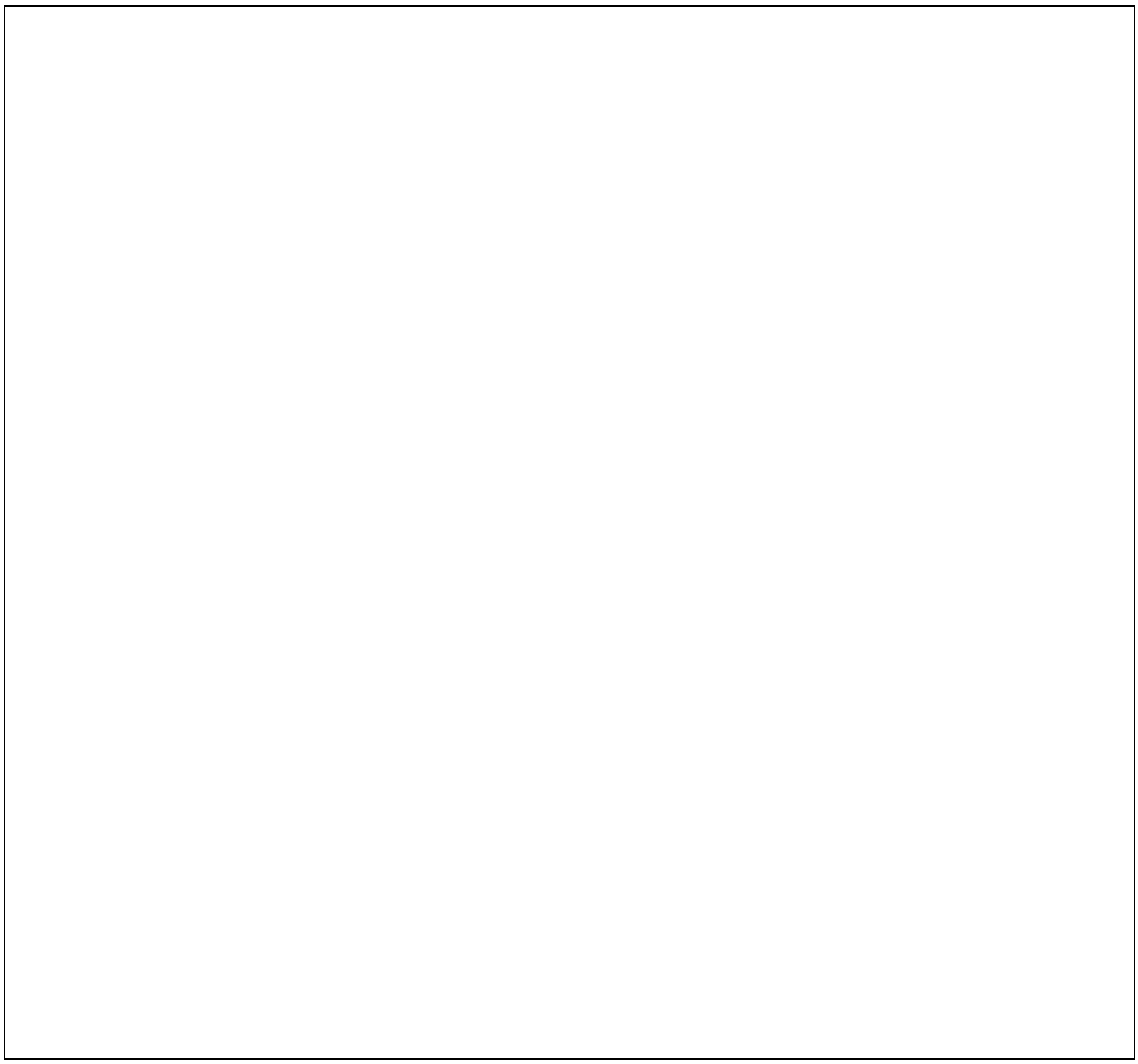
SUBMISSION:

SECTION 1 (ALGORITHM)

Provide a clear, efficient, and well-structured algorithm to the given case study problem.

- Start.
- Ask user for name.
- Show the available books to display with the names for Book 1, Book 2, and Book 3 with their prices.
- Prompt the user to enter a number for the quantity of books they wish to order, with the name of the books on display.
- If the user enters invalid inputs for book quantities, the user is prompted again. But if the total number of books are all zero or exceeds 10 the program closes.
- Display an **error** if input contains negative numbers, all books containing the input of zero, and invalid characters, or if a single quantity of book exceeds 10 books, or if total quantity of books exceeds 10.
- The discount is applied depending on numbers of books: 0% discount for 0 – 3, 5% discount for 4 – 6, 10% discount for 7 – 9, 15% discount for exactly 10 books.
- Display the 'Customer Order Summary' with the customer's name, their total number of books ordered, names of the books for Book 1, Book 2, and Book 3 with their quantity and corresponding price, and display total cost without discount, with the applied discount for total cost, and the discount number (%) that changes depending on number of books.
- End.

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SECTION 2 (COMPLETE JAVA CODE)

Provide a clear, structured, and working solution to the given case study problem.

Write your Java code in the space provided below, ensuring it is well-indented and readable.

```
import java.util.Scanner;


public class bookPurchase {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);


        // Variables for book pricing, book quantities
        double[] bookPrice = {12.99, 15.49, 20.99};
        double discount = 0;
        int[] bookQuantity = new int[3];
        String[] bookNames = {"'Mystery of the Ancient Tomb' (Book 1)", "'The Rise of the Robots' (Book 2)", "'Understanding the Universe' (Book 3)"};
        String name;



        // Customer Name to accept user input
        System.out.print("Enter the customer's name: ");
        name = scanner.next();


        // Display available books
        System.out.println("\nAvailable Books:");
        System.out.println("\n1. Mystery of the Ancient Tomb - $" + bookPrice[0]);
```

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```
System.out.println("\n2. The Rise of the Robots - $" + bookPrice[1]);
System.out.println("\n3. Understanding the Universe - $" + bookPrice[2] + "\n");

// Increases index for both bookNames and bookQuantity per user input
for (int i = 0; i < 3; i++) {
   System.out.print("\nEnter quantity for " + bookNames[i] + ":  ");
   bookQuantity[i] = scanner.nextInt();

   // Repeats for statement until user enters valid input.
   while (bookQuantity[i] > 10 || bookQuantity[i] < 0) {

      // Gives error if number is greater than 10
      if (bookQuantity[i] > 10) {
         System.out.println("\nError! Quantity for book cannot exceed 10.");
      }

      // Gives error if number is less than 0
      else if (bookQuantity[i] < 0) {
         System.out.println("\nError! Requires positive numeric values.");
      }

      // Ask for input again
      System.out.print("\nEnter quantity for " + bookNames[i] + ":  ");
      bookQuantity[i] = scanner.nextInt();
   }
}
```

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```
    // Add up the book quantities in new variable
    int totalBooks = bookQuantity[0] + bookQuantity[1] + bookQuantity[2];



    // If all inputs for bookQuantity contains zero then it will display an error
    if (totalBooks == 0) {
       System.out.println("\nError! All books contain the input of zero.");
       return;
    }
       // If total book quantity is greater than 10, error is displayed
       else if (totalBooks > 10) {
       System.out.println("\nError! Total number of books exceed 10.");
       return;
    }

    //Total cost of all the books
    double total_cost = (bookQuantity[0] * bookPrice[0]) + (bookQuantity[1] * bookPrice[1]) +
(bookQuantity[2] * bookPrice[2]);


    //Change discount number based on totalBooks
    if (totalBooks <= 3) {
       discount = 0.0;
    } else if (totalBooks <= 6) {
       discount = 0.05;
    } else if (totalBooks <= 9) {
```

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```
            discount = 0.10;
        } else if (totalBooks == 10) {
            discount = 0.15;
        }


        // Print out the customer summary
        System.out.println("\n\nCustomer Order Summary: ");
        System.out.println("\nCustomer Name: " + name);
        System.out.println("\nMystery of the Ancient Tomb: " + bookQuantity[0] + " copies" + " @ " + "$" + bookPrice[0] + " each");
        System.out.println("\nThe Rise of the Robots: " + bookQuantity[1] + " copies" + " @ " + "$" + bookPrice[1] + " each");
        System.out.println("\nUnderstanding the Universe: " + bookQuantity[2] + " copies" + " @ " + "$" + bookPrice[2] + " each\n");
        System.out.printf("\nTotal Cost: $%.2f\n", total_cost);
        System.out.printf("\nDiscount Applied: %.1f%%\n", discount * 100);
        System.out.printf("\nAmount Payable: $%.2f\n", (total_cost - (total_cost * discount)));
    }
}
```

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SECTION 3: SCREENSHOT FOR PROGRAM EXECUTION (VALID INPUT)

```
Enter the customer's name: Alice
Available Books:
1. Mystery of the Ancient Tomb - $12.99
2. The Rise of the Robots - $15.49
3. Understanding the Universe - $20.99

Enter quantity for 'Mystery of the Ancient Tomb' (Book 1): 2
Enter quantity for 'The Rise of the Robots' (Book 2): 2
Enter quantity for 'Understanding the Universe' (Book 3): 3

Customer Order Summary:
Customer Name: Alice
Mystery of the Ancient Tomb: 2 copies @ $12.99 each
The Rise of the Robots: 2 copies @ $15.49 each
Understanding the Universe: 3 copies @ $20.99 each

Total Cost: $119.93
Discount Applied: 10.0%
Amount Payable: $107.94
```

```
Enter the customer's name: Joey
Available Books:
1. Mystery of the Ancient Tomb - $12.99
2. The Rise of the Robots - $15.49
3. Understanding the Universe - $20.99

Enter quantity for 'Mystery of the Ancient Tomb' (Book 1): 5
Enter quantity for 'The Rise of the Robots' (Book 2): 0
Enter quantity for 'Understanding the Universe' (Book 3): 5

Customer Order Summary:
Customer Name: Joey
Mystery of the Ancient Tomb: 5 copies @ $12.99 each
The Rise of the Robots: 0 copies @ $15.49 each
Understanding the Universe: 5 copies @ $20.99 each

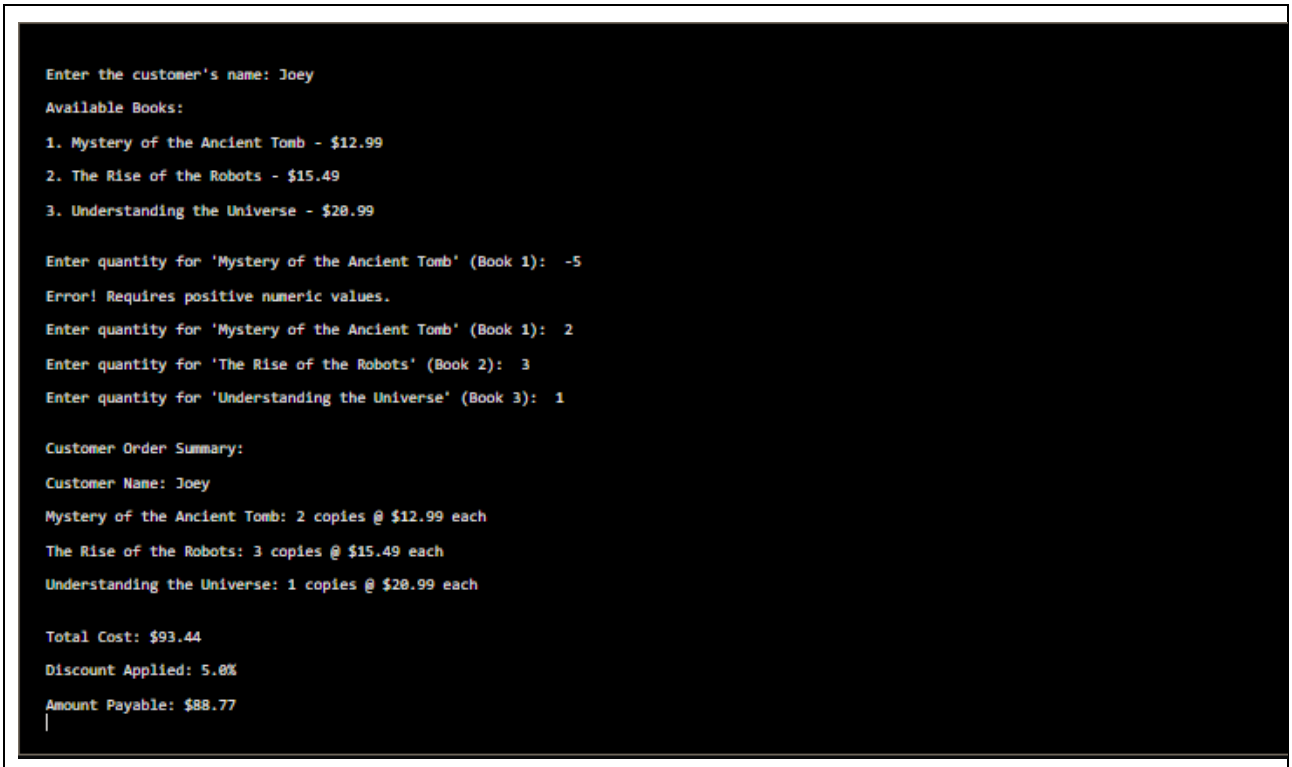
Total Cost: $169.90
Discount Applied: 15.0%
Amount Payable: $144.42
```

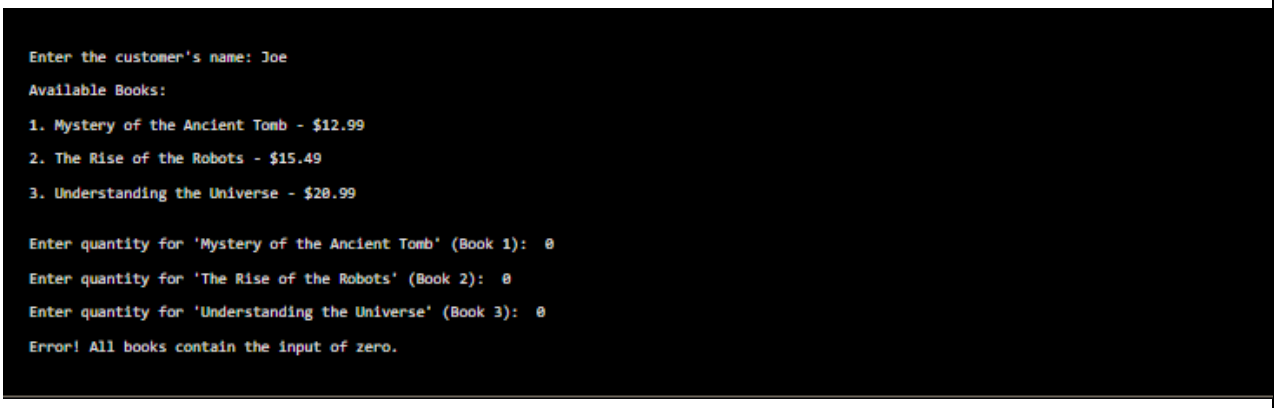
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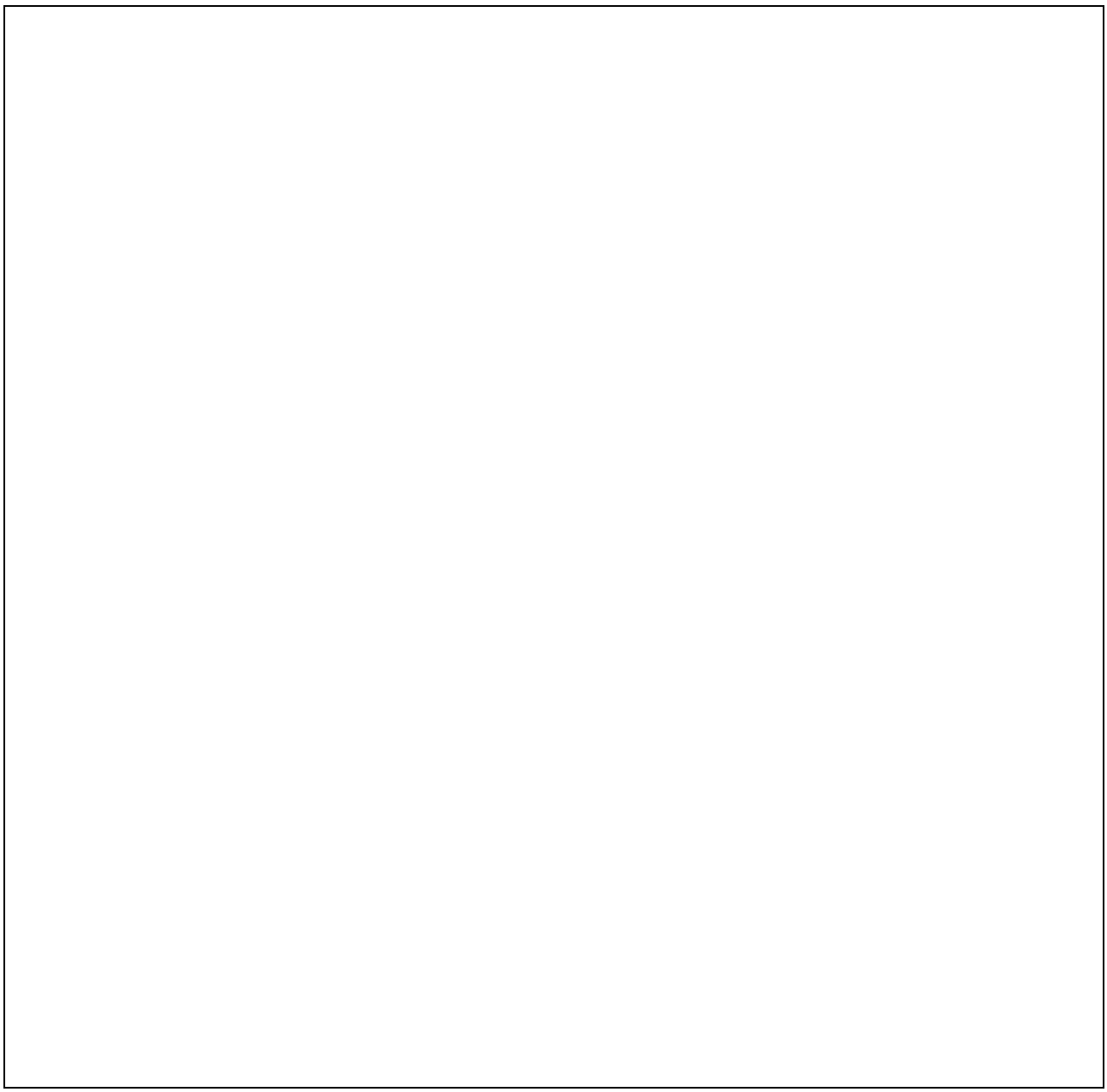
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SECTION 4: SCREENSHOT FOR PROGRAM EXECUTION (INVALID INPUT)





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